

Institute of Actuaries of India

**Subject ST6 – Finance and
Investment B**

April 2016 Examination

INDICATIVE SOLUTION

Solution 1:

The following clues shall enable you to match the prices to the relevant contracts:

- The strike rates under (a) and (b) are very similar. But, the rate for (a) is 4% *pa* quarterly compounded corresponds to 3.969% continuously-compounded, whereas the rate for (b) is 3% *pa* monthly compounded corresponding to 3.990% *pa* continuously-compounded. Hence, the price of (b) has to be slightly lower than the price of (a).
- (d) is deeply in-the-money and thus will have the highest price
- (c) is unlike (a) and (b) but similar to (d). In fact, (c) is also in-the-money, but level of moneyness is lesser than (d). Thus, (c) is likely to have a higher price than (a) and (b) but lower than the price of (d).
- The two strike prices for a collar are usually chosen to make the price zero with the two rates usually straddling the current interest rate. Thus the value of (e) is zero.
- Finally, a fixed – for – floating swap (*i.e.* pay fixed and receive floating) can be replicated by 1 long cap and 1 short floor (on the same interest rate). Thus, its price has to be the difference between the prices of (a) and (c). This means that the value of the swap should be price of (a) less the price of (c) = - Rs.10,000. Thus, in this instance the floating rate payer receives Rs. 10,000 which is therefore the value of (f).

Thus, the final values and the contracts match up as follows:

Contract	Price
A	Rs. 23,000
B	Rs. 22,000
C	Rs. 28,000
D	Rs. 40,000
E	Rs. 0
F	Rs. 10,000

[15 Marks]

Solution 2:

i)

Probability of Company X defaulting is 3.6%

Probability of Yes Bank defaulting is 0.5%

Given that each of the events are independent of each other, and that Kotak Bank will suffer a loss only if both these events happen together.

The probability of Kotak Bank incurring a loss is the product of the probability of default of company X and the probability of default of Yes Bank = 0.5% * 3.6% = 0.018%

[3]

ii)

The expected loss, given a \$100 million exposure with a probability of 0.018% is

$$\$100 \text{ million} * 0.018\% = \$18,000$$

[2]

[5 Marks]**Solution 3:**

The standard deviation of the original portfolio is:

$$\sqrt{\left(\left(\left(\frac{100}{150}\right)^2 * (0.25)^2\right) + \left(\left(\frac{50}{150}\right)^2 * (0.2)^2\right) + \left(2 * \left(\frac{100}{150}\right) * \left(\frac{50}{150}\right) * 0.2 * 0.25 * 0.2\right)\right)} = 0.036667$$

667

Thus, the daily volatility is

$$\frac{0.036667}{\sqrt{250}} = 0.012111$$

Hence, the 99% Value at Risk number is Rs. 150 million * 2.33 * 0.012111 = Rs. 4.232655 million

The standard deviation of the altered portfolio is:

$$\sqrt{\left(\left(\left(\frac{50}{150}\right)^2 * (0.25)^2\right) + \left(\left(\frac{100}{150}\right)^2 * (0.2)^2\right) + \left(2 * \left(\frac{100}{150}\right) * \left(\frac{50}{150}\right) * 0.2 * 0.25 * 0.2\right)\right)} = 0.029167$$

167

Thus, the daily volatility is

$$\frac{0.029167}{\sqrt{250}} = 0.010801$$

Hence, the 99% Value at Risk number is Rs. 150 million * 2.33 * 0.010801 = Rs. 3.775031 million

Thus, the change in the VaR is [Rs. 4.232655 million – Rs. 3.775031 million] = Rs. 0.457624 million

[7 Marks]

Solution 4:

i)

S_t follows Geometric Brownian Motion. This implies

$$S_t = S_0 \exp(\mu t + \sigma W_t).$$

Let us define $L_t = \ln(Z_t)$. Then

$$L_t = \ln(B_t^{-1} S_t) = \ln(B_t^{-1}) + \ln(S_t) = \ln(e^{-rt}) + \ln(S_0 e^{\mu t + \sigma W_t}) = -rt + \mu t + \sigma W_t.$$

Hence,

$$dL_t = (\mu - r)dt + \sigma dW_t.$$

Now as $Z_t = f(L_t) = \exp(L_t)$, one can apply Ito's formula to get:

$$dZ_t = \left[\left((\mu - r)f'(L_t) + \frac{1}{2}\sigma^2 f''(L_t) \right) dt + \left(\sigma f'(L_t) \right) dW_t \right]$$

Substituting $f'(L_t) = f''(L_t) = \exp(L_t) = Z_t$, one gets

$$dZ_t = \left(\mu - r + \frac{\sigma^2}{2} \right) Z_t dt + \sigma Z_t dW_t.$$

[6]

ii)

A portfolio is self-financing if and only if changes in its value depend only on changes in the prices of the assets constituting the portfolio.

Mathematically: If V_t denotes the value of the portfolio (ϕ_t, ψ_t) , then the portfolio is self-financing if and only if

$$dV_t = \phi_t dS_t + \psi_t dB_t$$

A replicating strategy for X is a strategy which involves investing in specifiable quantities (ϕ_t, ψ_t) of stock and risk free bonds, such that the portfolio of (ϕ_t, ψ_t) of stocks and bonds will be self-financing the portfolio (ϕ_t, ψ_t) and will have terminal value equal to the magnitude of the claim; i.e. $V_T = \phi_T S_T + \psi_T B_T = X$.

This means that the portfolio's cash flows at the claim exercise date match the cash flows under the claim.

When the underlying stock follows a continuous geometric Brownian motion process, there is an additional technical constraint for the strategy to work; namely:

$$\int_0^T \phi_t^2 \sigma^2 dt < \infty.$$

[6]

iii)

We have from (a)

$$Z_t = \left(\mu - r + \frac{\sigma^2}{2} \right) Z_t dt + \sigma Z_t dW_t.$$

Notice, the drift of this process is

$$\left(\mu - r + \frac{\sigma^2}{2} \right) \geq 0.$$

Since, the process is not driftless, it will not be a Martingale.

We can thus use the Cameron – Martin – Girsanov (CMG) theorem to convert the process into a Martingale. The CMG theorem states that there exists a probability measure Q , equivalent to the measure P (defined by the probability distribution of W_t), such that Z is a Martingale.

Thus, to apply the theorem we set

$$\gamma_t = \left(\mu - r + \frac{\sigma^2}{2} \right)$$

and then verify that γ_t is a pre-visible process and that

$$E_P \left[\exp \left(\frac{1}{2} \int_0^T \gamma_t^2 dt \right) \right] < \infty.$$

Using the CMG theorem, the measure Q , that is equivalent to P is such that

$$\frac{dQ}{dP} = \exp \left(-\frac{1}{2} \int_0^T \gamma_t^2 dt - \int_0^T \gamma_t dW_t \right)$$

and the Brownian motion

$$\widetilde{W}_t = W_t + \int_0^t \gamma_s ds$$

is a Q -measure Brownian motion.

Alternately, this can also be written as

$$d\widetilde{W}_t = \gamma dt + dW_t.$$

If one substitutes dW_t into the SDE for Z , one would get,

$$dZ_t = \sigma Z_t d\widetilde{W}_t.$$

This is the SDE for a driftless process under measure Q . Hence, Z is a Martingale under measure Q .

The next step in the construction of the replication strategy is to form the discounted expected claim process

$$E_t = E_Q(B_t^{-1}X | \mathcal{F}_t)$$

and show that this is a Q -measure Martingale as well. This can be done in a manner similar to the way Z_t is shown to be a Q -Martingale.

As both Z_t and E_t are Q -Martingales, the Martingale Representation Theorem (MRT) gives us a pre-visible process ϕ_t such that $dE_t = \phi_t dZ_t$.

It may be noted that in order to Note that in order to apply the MRT we need to show that both Z_t and E_t are Q -Martingales and that the volatility of Z_t is non zero with probability 1.

The replication strategy then consists of holding ϕ_t units of stock and ψ_t risk free bonds, where ψ_t is given by

$$\psi_t = E_t - \phi_t Z_t$$

[8]

iv)

The portfolio replicates the claim because the portfolio is self – financing and, at time T , when the claim falls due, the portfolios proceeds are

$$\begin{aligned} \phi_T S_T + \psi_T B_T &= \phi_T S_T + (E_T - \phi_T Z_T) B_T = \phi_T S_T + (E_T - \phi_T B_T^{-1} S_T) B_T = E_T B_T \\ &= E[B_T^{-1} X | \mathcal{F}_T] B_T = X \end{aligned}$$

i.e. the portfolio's proceeds will match the claim amount.

The value of the portfolio at any time t can be written

$$V_t = \phi_t S_{Tt} + \psi_t B_t = B_t E_t.$$

The SDE for V_t is given by

$$dV_t = d(B_t E_t) = B_t dE_t + E_t dB_t.$$

To show that the portfolio is self-financing, one needs to show that

$$dV_t = \phi_t dS_t + \psi_t dB_t$$

As the portfolio replicates the claim, the arbitrage-free condition requires that the value of the claim equals the value of the replicating strategy. Therefore, either of the above two SDEs gives the stochastic differential equation for the value of the claim.

[3]

[23 Marks]

Solution 5:

i) NSE put option

Input Data

Stock Price now (P)	7000
Exercise Price of Option (EX)	6650
Number of periods to Exercise in years (t)	0.5
Compounded Risk-Free Interest Rate (rf)	7.00%
Standard Deviation (annualized σ)	25.00%

Output Data

Present Value of Exercise Price (PV(EX))	6421.2760
e^{-rt}	0.1768
d1	0.5765
d2	0.3998
Delta N(d1) Normal Cumulative Density	0.7179

Function	
Bank Loan $N(d2) \cdot PV(EX)$	4208.0769
<i>Value of Put</i>	
	238.3160
<i>Value</i>	
	500,000,000
Number of option required	
	71,429
Total Cost	
	17,022,573

Brazilian put option

Input Data

Stock Price now (P)	40,000
Exercise Price of Option (EX)	38000
Number of periods to Exercise in years (t)	0.5
Compounded Risk-Free Interest Rate (rf)	10.00%
Standard Deviation (annualized σ)	35.00%

Output Data

Present Value of Exercise Price (PV(EX))	36,147
$\sigma \cdot t^{0.5}$	0.2475
d1	0.5330
d2	0.2855
Delta $N(d1)$ Normal Cumulative Density	0.7030

Function	
Bank Loan $N(d2) \cdot PV(EX)$	22136
<i>Value of Put</i>	2130.7259
Exposure	
	500,000,000
Number of option required	
	13,158
Option Price	
	28,035,867

FX option

Spot rate (Real/Rs)	20
Strike price (same units as Spot)	19
volatility (annualized)	15.00%
domestic interest rate (annualized)	7.00%
foreign interest rate (annualized)	10.00%
time to maturity in days	180
time to maturity in years	0.50
Required Expressions	
Black-Scholes D1	0.3952
Black-Scholes D2	0.2891
Put Option	

Option Prices	0.4970
Exposure	
	500,000,000
number of contract required	25,000,000
Price	12,425,532.00

[10]

ii) Correlation Matrix

	Indian	Brazil	FX
Indian	100%	50%	0%
Brazil	50%	100%	0%
FX	0%	0%	100%

Total stress = $3 \times .05 \times 50,00,00,000 = 7,50,00,000$

Stress after diversification = 5,00,00,000

Diversification = 67%

Changes to individual stress $5 / .67 = 8\%$

[5]

iii) New NSE put option

Input Data

Stock Price now (P)	7000
Exercise Price of Option (EX)	6440
Number of periods to Exercise in years (t)	0.5
Compounded Risk-Free Interest Rate (rf)	7.00%
Standard Deviation (annualized ??)	25.00%

Output Data

Present Value of Exercise Price (PV(EX))	6218.4989
$\sigma \cdot t^{0.5}$	0.1768
d1	0.7581
d2	0.5813
Delta N(d1) Normal Cumulative Density Function	0.7758
Bank Loan $N(d2) \cdot PV(EX)$	4474.0473
<i>Value of Put</i>	174.9902
Value	500,000,000
Number of option required	71,429
	12,499,301

New Brazilian put option

Input Data

Stock Price now (P)	40,000
Exercise Price of Option (EX)	36800
Number of periods to Exercise in years (t)	0.5
Compounded Risk-Free Interest Rate (rf)	10.00%
Standard Deviation (annualized σ)	35.00%

Output Data

Present Value of Exercise Price (PV(EX))	35,005
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$\sigma \cdot t^{0.5}$	0.2475
d1	0.6627
d2	0.4152
Delta N(d1) Normal Cumulative Density Function	0.7462
Bank Loan $N(d2) \cdot PV(EX)$	23139
<i>Value of Put</i>	1716.0826
Exposure	500,000,000
Number of option required	13,587
Option Price	23,316,339

New FX option

Inputs	
Spot rate (Real/Rs)	20
Strike price (same units as Spot)	18.4
volatility (annualized)	15.00%
domestic interest rate (annualized)	7.00%
foreign interest rate (annualized)	10.00%
time to maturity in days	180
time to maturity in years	0.50

Required Expressions

Black-Scholes D1	0.6977
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Black-Scholes D2	0.5917
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Put Option

Option Prices	0.3054
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Exposure

500,000,000

number of contract required

25,000,000

Price

7,635,372.36

Cost after	After	Before
NSE	12,499,301	17,022,573
IBSP	23,316,339	28,035,867
FX	7,635,372	12,425,532
	43,451,013	57,483,972

[8]

[23 Marks]**Solution 6:****i)**

If the gamma of an option position is large and negative the writer will suffer a significant loss for large up or down market movement.

[3]

ii)

Differentiate black scholes equation with respect to share price twice. (Further work is required by typing the equation)

Again further work is required by typing the equation but proof is obtain by constructing the equation for gamma and delta and estimating the time and weight of the option.

[5]

[8 Marks]

Solution 7:

i)

The riskiness of the bond is determined by the expected default by the bond. The price of the bond would be due to both the default probability and illiquidity premium of holding the bond specially during the stressed market condition.

[3]

ii)

The first capital approach focuses only on the default of the bond. This approach is useful for the life insurance companies which are more inclined to hold the bonds till maturity.

The second approach takes in account of the market movement in the price of the bonds due to spreads. This capital requirement is more suitable for companies which actively trade the corporate bonds.

[3]

iii)

The bond is priced at par thus the GRY pre default is the bonds is 5%

Default	2%						
Coupon rate	5%						
Principle	10,000.00						
Recovery rate	50%					3.90%	5%
	Start year	Default %	Bond at the End	recovery	coupon	Cashflow	
31/12/2015						-10000	-
							10000
31/12/2016	10000	2%	9800	100	490	590	500
31/12/2017	9800	2%	9604	98	480.2	578.2	500
31/12/2018	9604	2%	9411.92	96.04	470.596	9978.556	10500

		Stressed value			9,707		
	Start year	Default %	Bond at the End	recovery	coupon	Cashflow	
	31/12/2015					-10000	
1	31/12/2016	10000	4%	9600	200	480	680 654.4997
2	31/12/2017	9600	4%	9216	192	460.8	652.8 604.7574
3	31/12/2018	9216	4%	8847.36	184.32	442.368	9474.048 8447.673

The capital requirement is $10000 - 9707 = 293.00$

The second approach $10000 - 9476 = 524.00$

[5+4+4=13]

[19 Marks]
